

Thomas G. Wildeboer

Quantitative and Computational Biologist

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Education

Doctor of Philosophy – Ecology and Evolutionary Biology 2023 – Current

University of Toronto

Topic: mathematical and computational modeling of complex trait evolution

Supervisors: Jacqueline Sztepanacz, Tomomi Parins-Fukuchi

Honours Bachelor of Science (distinction) – Genome Biology, Ecology and Evolutionary Biology 2019 – 2023

University of Toronto-St. George, St. Michael's College

Honours title: Multivariate comparisons reveal when **P** is an adequate substitute for **G**

Honours supervisor: Jacqueline Sztepanacz

Work & Teaching Experience

Graduate Research Assistant – Sztepanacz & Parins-Fukuchi Labs, University of Toronto 2023 – Current

Toronto, Ontario

- Developed Julia framework for simulations of complex trait evolution.
- Developed mathematical models of evolution and extinction in stochastic environments.

Teaching Assistant – University of Toronto 2023 – Current

Toronto, Ontario

- EEB125 – Computation and Data Science for the Life and Physical Science
- EEB225 – Biostatistics for Biological Sciences
- BIO220 – From Genomes to Ecosystems in a Changing World
- BIO120 – Adaptation and Biodiversity

Software Developer – Whispering Pines Landscaping Inc. Summers 2019 – 2023

Orangeville, Ontario

- Cloud-based inventory management software development in C#.

Laboratory Assistant – Mahler Lab, University of Toronto Summer 2021

Toronto, Ontario

- Assistance with computational research investigating the evolution of morphological diversity among tropical lizards.

Software Developer – Maple Reinders Group Summer 2018

Mississauga, Ontario

- Enterprise systems development in C#.

Additional Experience

Student – Evolutionary Quantitative Genetics Workshop Jun 2025

Mountain Lake Biological Station, Virginia

Publications

- **Wildeboer T**, Parins-Fukuchi T, Sztepanacz J. Genetic and demographic constraints on evolution as a unifying explanation for stasis and predictability. In prep for *Evolution*.
- **Wildeboer T**, MacKenzie A, Sztepanacz J. A multivariate comparison of phenotypic and genetic covariance matrices: when **P** can be used for making evolutionary predictions. In revision at *Evolution* (EVO-25-0396).

Conference Presentations

- **Wildeboer T**, Parins-Fukuchi T, Sztepanacz J. 2025. The effect of pleiotropy on extinction risk over macroevolutionary time. Atwood Colloquium in Ecology and Evolution, Toronto, Ontario.

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- **Wildeboer T**, Sztepanacz J. 2024. Multivariate comparisons reveal when **P** is an adequate substitute for **G**. (Symposium: Predicting evolutionary responses to a changing world) Third Joint Congress on Evolutionary Biology, Montréal, Québec.
- **Wildeboer T**, Sztepanacz J. 2023. Multivariate comparisons reveal when **P** is an adequate substitute for **G**. (Poster) EEB Undergraduate Research Fair, Toronto, Ontario.

Selected Projects

Computing Agent-Based Models

- A collection of agent-based models for simulating dynamical systems and evolution with Python and NumPy.

RNA-seq in *Drosophila suzukii*

- Snakemake pipeline to reanalyze Deng et al. (2022) RNA-seq data to test for sex-biased gene expression in the spotted-wing fruit fly with DESeq2.

Phylogenetic Inference with Quantitative Traits

- Julia simulations of quantitative trait evolution under genetic constraints with analysis of topological distance between true and inferred phylogenies under various optimality criteria.

Community Involvement

Judge – Undergraduate Engineering Research Day, Faculty of Engineering, University of Toronto <i>Toronto, Ontario</i>	Aug 2026
Guest Lecturer – EEB125, Computation and Data Science <i>Toronto, Ontario</i>	Mar 2026
Panelist – St. Michael's College Postgrad Primer, Research Edition <i>Toronto, Ontario</i>	Jan 2026
Grad-Undergrad Mentor – EEB Graduate Student Association <i>Toronto, Ontario</i>	2025-2026
Graduate Peer Mentor – EEB Graduate Student Association <i>Toronto, Ontario</i>	2025-2026
Panelist – Graduate School Application Info Session, EEB <i>Toronto, Ontario</i>	Oct 2025
Volunteer – Atwood Colloquium in Ecology and Evolution <i>Toronto, Ontario</i>	Apr 2025
Panelist – EEB Alumni Panel <i>Toronto, Ontario</i>	Feb 2025
Journal Club Chair – EEB Graduate Student Association <i>Toronto, Ontario</i>	2024-2025
Science Fair Judge – Science Rendezvous <i>Toronto, Ontario</i>	May 2024

Last updated Aug 18, 2026